FE5108 Class Project · Team Plan

Portfolio Theory and Investments — Practitioner Segment Class Project (Fall 2026)

Deadline: **Sunday, 8 Nov 2026**, by email | Text ≤ 8 pages (exhibits excluded) | Code as attachment or appendix | Cover page states each member's contribution

1. Division of work

Each member owns one Part end to end: **data loading, code, exhibits and the write-up** for that Part, working independently without waiting for anyone else's data. Members are numbered 1–5 (Member 1 owns Part 1, and so on) to avoid confusion with question numbers such as 1A or 2B.

Member	Part	Points	Page budget	Data to load
Member 1 ____	Part 1 Factor investing (1A–1E)	22	≈ 1.75 pages	File 1: FF 3 factors, momentum portfolios, beta portfolios, 5 MSCI indexes
Member 2 ____	Part 2 Security-level data (2A–2D)	20	≈ 1.5 pages	File 2: stock-level data; File 1: HML, MSCI USA
Member 3 ____	Part 3 Portfolio construction (3A–3E)	21	≈ 1.75 pages	File 3: 49 industries; File 1: MSCI USA
Member 4 ____	Part 4 (to be released)	≈18	≈ 1.5 pages	TBD
Member 5 ____	Part 5 (to be released)	≈19	≈ 1.5 pages	TBD (File 1 is labelled “parts 1 and 5”, so Part 5 will likely use it)

Note: Parts 1–3 total 63 points; the remaining ~37 points are assumed to be split between Parts 4 and 5. Page budgets are proportional to points and sum to 8 pages; adjust once Parts 4/5 are released.

2. Data files: all provided by the professor

All data for Parts 1–3 is in the 3 files provided on the class website — **no need to download anything from the Fama-French or MSCI websites**. The files sit in `data/raw/` of the repo; everyone reads what they need directly, so **there are no data hand-offs between members**.

File	Contents	Read by
File 1 class project data (FF MSCI) (parts 1 and 5).xlsx	4 sheets: · FF - portfolios sorted on beta · FF - portfolios sorted on pmom (momentum) · FF - 3 factor returns · MSCI - factor totret index (5 index levels)	Member 1 (all) Member 2 (HML; MSCI USA) Member 3 (MSCI USA) Member 5 (likely, for Part 5)
File 2 class project data (part 2 - factor portfolio construction).csv	mlfactor stock-level data: stock_id, date, Pb, Bv, R1M_Usd, Vol3Y_Usd, Mkt_Cap_3M_Usd; Nov 1998 – Mar 2019	Member 2
File 3 class project data (part 3 - pf construction).xlsx	FF 49 industry portfolios, value-weighted monthly returns, already trimmed to Jan 1995 – Jul 2023	Member 3

Things to watch out for when loading:

Data	Caveat	Member
FF beta portfolios	In the header, the 8 columns between Lo 10 and Hi 10 were converted by Excel into dates (2023-12-02 … 2023-12-09); they are actually deciles 2–9. Data start only in Jul 1963.	1
FF momentum & beta	Several blocks are stacked in one sheet (VW monthly, EW monthly, annual, number of firms, …). Use only the “Value Weighted Returns -- Monthly” block.	1
FF 3 factors	Annual factors follow below the monthly block — don't read them in. Units are percent: divide by 100.	1, 2
MSCI indexes	Index levels, not returns: compute month-end to month-end % change. Numbers contain thousands separators (e.g. 1,182.011) — convert to numeric first. Series starts 30 Dec 1994, so the first return is Jan 1995.	1, 2, 3
Stock-level data	Percentiles are on a 0–1 scale (the assignment says 0–100; same meaning). R1M_Usd is a decimal, one-month-forward return: shift by one month before comparing with MSCI.	2
49 industries	No -99.99 missing values in Jan 1995 – Jul 2023; ready to use. Units are percent.	3

The one thing to coordinate: shared data must give identical numbers

- MSCI USA monthly returns are computed separately by Members 1, 2 and 3; FF HML is read separately by Members 1 and 2.
- After loading, compare a few values in the group chat (e.g. MSCI USA return and HML for Jan 1995) so the numbers in the report stay consistent.
- Wording: Member 2's discussion of how HML is constructed (2C/2D) should match Member 1's description in 1A — check each other's text.

3. Team conventions

All conventions are implemented in `src/config.py` and `src/utils.py` — use them instead of writing your own. Add these lines at the top of every notebook in `partN/`:

```
import sys; sys.path.append("..")
from src import config, utils
config.apply_style()
```

File	What it provides
src/config.py	Paths and sheet names of the 3 data files; all analysis periods from the assignment (config.PERIODS, e.g. "1C", "3E_covid"); chart sizes, font sizes and colour palette; apply_style()
src/utils.py	Units & dates: pct_to_decimal(), levels_to_returns(), to_monthly_index(), select_period() Statistics: annualized_return(), annualized_volatility(), sharpe_ratio(), tracking_error(), t_stat_mean(), cumulative_value() Output: new_fig(), percent_axis(), save_fig(), save_table()

- **Units:** all returns in **decimals** (0.01 = 1%) inside code; convert to percent only when displaying. FF percent data → `pct_to_decimal()`; MSCI index levels → `levels_to_returns()` (also strips thousands separators).
- **Dates:** monthly Period index; each row is the return earned in that month → `to_monthly_index()`. Slice periods with `select_period(df, "1C")` so nobody cuts the sample

differently.

- **Volatility:** always std(monthly) × √12.
- **Annualized return:** the method must be stated explicitly — `method="arithmetic"` (mean × 12) or `method="geometric"` (compound growth). There is no default: the assignment asks us to think about which mean is appropriate (1A, 2C), so choose per question and say so in the report.
- **Sharpe ratio:** follows footnote 2 of the assignment — (annualized return − annualized risk-free return) / annualized volatility.
- **Exhibit size:** create figures at their **printed size** with `new_fig("full")` (6.0 × 2.8 in) or `new_fig("half")` (3.0 × 2.3 in, two side by side). Never resize them in Word — this keeps fonts readable and stops labels overlapping.
- **Exhibit style:** small fonts (9 pt title, 8 pt labels, 7 pt ticks and legend, 6.5 pt source note); colour-blind-safe palette; market/benchmark always grey; top-5 green / bottom-5 red preset for 3A.
- **Exhibit content:** every chart/table states what is shown, data source, period and investable universe; all axes labelled. `save_fig(fig, name, source="...")` prints the source line under the chart.
- **File names:** include the question number, e.g. `fig_1C_sharpe.png`, `tab_3A_weights.csv`, so merging the report is easy.